Validation and AI Use — Project 1 (ACI)

Content-equivalent working source for `Validation-and-AI-Use.pdf`.

Decision, version, and as-of boundary

- **Decision/user:** initiate, watch/defer, or do not initiate Albertsons Companies, Inc. (NYSE: ACI); fund investment committee with no position. **Recommendation:** watch/defer, $13–$19 per share.
- **Repository:** github.com/mpuntillo33/ticker-research, branch `PROJECT-1`, frozen at tag `p1-final` (commit recorded in the manifest).
- **Bridge date:** Sept 12, 2026. **Price:** $13.09, close Oct 16, 2026. **Source cutoff:** information available by Oct 19, 2026.
- **Environment:** Windows 11, Python 3.12.10, packages in `requirements.txt`; engine uses the standard library only.

Data and convention ledger

Material input	Source	As-of/availability	Definition/units	Transformation/exclusion	Risk
FY2025 revenue $83,172.5m; gross profit $22,606.7m; SG&A $21,891.3m	FY2025 Form 10-K (acc. 0001646972-26-000032), statement of operations	53 wks to Feb 28, 2026; filed Apr 27, 2026	USD m	Revenue less $1,360m 53rd week; SG&A less $773.8m opioid charge (93.4% of GP)	Extra-week estimate is management's
Balance sheet: inventory $5,173.9m, PP&E $9,903.7m, cash $198.6m, debt incl. finance leases $8,946.6m, equity $1,836.2m	10-K balance sheet	Feb 28, 2026	USD m	Opening balance sheet; "other" lines aggregated; balances to $26,765.9m	Aggregation hides line detail
Accounts payable $4,021.2m	10-K balance sheet (screenshot `docs/evidence/ap-10k-check.png`)	Feb 28, 2026	USD m	24.2 days of FY2025 cost of sales	Days held constant
Opioid Settlement Framework $773.8m	10-K Note 12: $655.1m + $21.7m + ≈$97.0m	Filed Apr 27, 2026	USD m, undiscounted, pre-tax	Debt-like bridge item	Overstates slightly (undiscounted)
Q1 FY2026: identical sales −0.8%, adj. EBITDA $1,013.0m; cover shares 485,325,683 (Jul 24, 2026)	Q1 FY2026 Form 10-Q	16 wks to Jun 20, 2026; filed Jul 28, 2026	% / USD m / shares	EBITDA run-rate check; market-cap shares	—
Q2 FY2026: identical sales −0.8%, adj. EBITDA $784.0m (4.1%), debt incl. finance leases $9,328.2m, cash $282.0m, diluted shares 488.2m; guidance withdrawn	Q2 FY2026 earnings release, Form 8-K Ex. 99.1	12 wks to Sept 12, 2026; filed Oct 13, 2026	USD m / m shares	Bridge and share basis	Q2 10-Q not yet filed (checked EDGAR Oct 19)
ACI price $13.09	stockanalysis.com (S&P Global data)	Close Oct 16, 2026, 4:00 PM EDT	USD	Market cap, reverse DCF	Single provider; Digrin showed $13.18 undated
10-yr Treasury 5.10%	FRED DGS10	Oct 15, 2026	%	Risk-free rate	—
Unlevered beta 0.85	Damodaran, Retail (Grocery and Food), cash-corrected	Jan 2026 dataset	—	Relevered at D/E 1.47 → 1.78	Industry, not company, beta
ERP 4.14%	Damodaran implied ERP (as used in Lab 06)	Sep 2026	%	CAPM	—
Peer market cap, debt, leases, cash, EBITDA TTM (ACI, KR, SFM)	stockanalysis.com statistics and quarterly balance sheets	Retrieved Oct 19, 2026; balance sheets Jun 20 / Aug 15 / Jun 28, 2026	USD m	EV = mkt cap + debt excl. all leases − cash	Provider EBITDA definition not published on page

Validation register

Claim/output	Failure mode	Test and expected result/direction	Actual result	Disposition	Evidence location
DCF engine is correct	Discounting/terminal error	**Known answer:** Lab 05 training case → $27.4974	$27.4974	Pass	`tests/test_engine.py`
Pro-forma method is correct	Statement linkage error	**Known answer:** Lab 09 ABG case → $291.75	$291.75	Pass	`tests/test_engine.py`
Statements articulate	Plugged cash, imbalance	BS balances and CF ties to cash every year, at every driver band extreme	Max gap 0.000000	Pass	`tests/test_engine.py`; report checks
Base year reconciles to filing	Silent normalization	Normalized revenue + $1,360m = $83,172.5m; SG&A/GP + charge = filed SG&A; opening BS balances	All reconcile	Pass	`analysis.base_year_reconciliation()`
DCF boundary/monotonicity	Wrong sign or invalid TV	WACC↑ → value↓; g↑ → value↑; g ≥ WACC refused	As expected; ValueError raised	Pass	`tests/test_engine.py`
Bridge	Double count or omission	EV − debt − opioid + cash = equity; +$100m debt → −$100m/488.2m per share	Exact	Pass	`tests/test_engine.py`
Independent source/definition check	Provider label ≠ filing definition	Provider "Total debt" for ACI vs bridge debt	Provider $15.92bn includes operating-lease liabilities; filing $9,328.2m = debt incl. finance leases. Bridge keeps the filing figure because FCFF is after rent	Pass (label reconciled)	stockanalysis.com ACI statistics, Oct 19, 2026; Q2 release
Directional changed-input test (written before run)	Margin not flowing to value	Lower FY2030 gross margin −0.5 pt → FY2030 EBIT↓, FCFF↓, value↓, revenue unchanged	As predicted	Pass	`tests/test_engine.py`
Peer policy and multiple definition	Mixed definitions	Same EV definition, same provider, leases excluded for all; ACI round-trip	Consistent	Pass	`comps.py`; Lab 08 policy (written before selection)
FY2026 forecast plausible	Forecast detached from current run-rate	Model FY2026 EBITDA within ±15% of annualized H1 FY2026 adj. EBITDA	−3.4%	Pass	report checks

Claim/output	Failure mode	Test and expected result/direction	Actual result	Disposition	Evidence location
Highest-risk claim: base value above price	Value depends on unproven cost recovery	Hold FY2030 SG&A/GP at FY2026's 94.0%	$12.69 < $13.09 price: the upside disappears	**Claim qualified:** recommendation set to watch/defer, initiation tied to SG&A/GP evidence	`tools/changed_input.py` `sga_to_gp_long 0.94`; Video 2
Checks fail loudly	Silent nonsense	SG&A/GP 97%, capex $2,400m → "Drivers inside historical band" FAILS	Fails (red in panel)	Pass	`tests/test_engine.py`; Video 3
Sensitivity ranking	Ranking presented as fact	Re-rank with narrow bands for the top three drivers	Leader changes from SG&A/GP to WACC	Ranking reported as range-conditional	`outputs/report.md`
Cold run	README insufficient	Fresh clone + new venv following README	First failure: `cd $HOME\Desktop` (OneDrive path); README steps then ran: 10/10 tests, $17.89	Pass with logged failure	`docs/cold-run-log.md`; recordings

Locked Changed-Input Record — human prediction first; AI closed during execution

Full record: `docs/locked-changed-input-record.md`.

Precommit before the run and before AI assistance on the prediction

Precommit timestamp	Repository commit	Material input	Old → new value/units	Expected output direction	Expected decision effect
Oct 19, 2026, 7:40 PM EDT (pushed 8:00 PM)	`7ae0841`	FY2026 gross margin	26.80% → 27.30%	Value up ≈ +$1.40/share; FY2030 EBIT and FCFF up	Likely to change watch/defer

Preserved execution result

Before/after output	Actual decision effect	Prediction reconciliation	Failure diagnosis or why action did not change	Evidence path
$19.474 → $19.483 (+$0.01); FY2030 EBIT unchanged; FY2030 FCFF −$2.0m (run 8:07:55 PM)	None	Missed (direction right, size far off)	In Max's words: see the record. Engine note: run on the engine at `7ae0841`, before the opioid obligation was added; that item shifts both values by a constant −$1.59/share	`docs/locked-changed-input-record.md`, commits `7ae0841` → `6d9c64e`

Max asked Claude only what the form fields meant while writing the prediction; Claude declined to review or suggest answers.

Named AI-use register

Tool/model surface and exposed version	Interaction date	Material task	Output/claim used or considered	Independent check	Accept/modify/qualify/reject	Effect on decision
Claude (Anthropic), Claude app; configured model `claude-opus-5-5`	Oct 19, 2026	Review Lab 10/11 working-capital design	Proposed projecting accounts payable at days of cost of sales ($4,021.2m, ≈24.2 days) because growth was destroying value	Max opened the 10-K balance sheet and confirmed $4,021.2m and total assets $26,765.9m (`docs/evidence/ap-10k-check.png`)	**Accept**	Growth now adds value; base moved $16.05 → $17.55 at 8.0% WACC before the WACC rebuild

Tool/model surface and exposed version	Interaction date	Material task	Output/claim used or considered	Independent check	Accept/modify/qualify/reject	Effect on decision
Same	Oct 19, 2026	WACC inputs	Proposed rejecting the provider's raw beta (0.23) and relevering Damodaran's grocery beta (0.85)	Damodaran Jan 2026 table; Kroger–Albertsons deal pricing (announced 2022 at $34.10, blocked Dec 2024) explains the low raw beta; computed WACC 7.68% vs Lab 06's 8.0% judgment	**Qualify**: used, with industry-beta limitation stated	WACC 7.7%
Same	Oct 19, 2026	Build `analysis.py`, `comps.py`, `app.ipynb`, tests, report tool	Engine code, driver base values and bounds, cases	10 tests incl. course known answers; Max reviewed every driver in the panel and accepted the base values (7:29 PM)	**Accept** after tests and review	All model outputs
Same	Oct 19, 2026	Locate newest share count and filings	488.2m diluted shares (Q2 release); Q2 10-Q not yet filed	EDGAR full-text search, Oct 19, 2026	**Accept**	Replaced 547.2m and 499.5m
Same	Oct 19, 2026	Draft memo, Edition B, this record; Video 2 script; video talking points	Draft text	Max reviewed and edited; figures trace to `outputs/report.md`; videos delivered in Max's voice	**Modify**	Communication only
ChatGPT / Codex (OpenAI)	Sept 3 – Oct 1, 2026	Labs 05–12 assistance and the Sept 3 report draft (disclosed in each lab)	Lab code and drafts	Lab known answers and partner reviews	As recorded in each lab	Lab inputs reused here

Pre-Edition-A note: a `/Finance` agent session record dated Aug 27, 2026 (Week 1, the disclosed AI-assisted screening period) exists in my course folder; it records no project output. Edition A itself was written in the supervised AI-off Lab 03 on Sept 1, 2026.

Limitations, monitoring, and kill/escalation rules

- Terminal value is 79.7% of EV; value is dominated by FY2030 SG&A/GP and WACC.
- No management guidance after Oct 13, 2026; FY2026 is anchored to the H1 run-rate instead.
- Peer EBITDA is a provider field applied identically to all companies; its exact definition is not published.
- The opioid obligation is undiscounted and pre-tax.
- **Kill/escalation:** if the bridge stops reconciling, any articulation check fails, or a new filing changes debt, shares or the opioid obligation, re-run `tools/make_report.py` before any committee use. Monitoring triggers are in the memo.

Locked Changed-Input Record — Project 1 (ACI)

Rules: Max writes Part 1 by hand with every AI tool closed, commits and pushes it, and only then runs the change. The run is AI-closed. Part 2 is filled from the preserved output.

Part 1 — Prediction (written before the run, AI off)

- Precommit timestamp:2026-10-19 7:40 PM EDT
- Repository commit (hash of the commit that adds this Part 1):7ae0841
- Material input (driver name): Gross margin FY26
- Old value → new value (with units): 26.80% -> 27.30%
- Why this input matters (one sentence): This input matter because if a grocery store chain can improve their margins, there could be massive upside to stock price and to revenue.
- Expected direction of value per share: Up
- Expected direction of FY2030 EBIT and FY2030 FCFF: If the Gross margin FY26 grows by .50%, then: FY2030 EBIT = 1800.00, and FY2030 FCFF = 1250.00 (in millions)
- Expected rough size of the value change ($ per share): +$1.40
- Expected decision effect (does watch/defer change?): This change would most likely change the watch/defer because a margin improvement in the grocery industry could have huge upside for futture earnings and free cash flows, which would improve the stock priceby a great magnitude.
- Reasoning (how the change flows through the statements):The change would allow more revenue to be kept, which would boost FY2030's EBITDA, and would also in turn boost the FCFF, which would cause more cash to be available to investors after expenses are paid, because profits from higher margins are generated.
- Asked Claude to explain what the form fields mean; it declined to review or suggest answers.

Part 2 — Preserved execution result (fill after the run)

- Command run: `python tools/changed_input.py gross_margin_fy2026 0.273`
- Run timestamp: Run at 2026-10-19T20:07:55
- Output (paste the printed table):

```
  | gross_margin_fy2026: 0.268 -> 0.273
Output                        Before       After      Change
Value per share ($)           19.474      19.483      +0.010
Enterprise value ($m)     18,553.188  18,557.967      +4.779
FY2030 revenue ($m)       85,547.423  85,547.423      +0.000
FY2030 EBIT ($m)           1,616.846   1,616.846      +0.000
FY2030 FCFF ($m)           1,151.181   1,149.170      -2.011
Terminal value share           0.797       0.796      -0.002
Market price for reference: $13.09 (2026-10-16)
Checks after change: all pass
```

- Actual direction and size: + $0.01
- Actual decision effect: No decision effect with this change.
- Prediction reconciliation (matched / partly / missed, and where): Missed
- Failure diagnosis, or why the action did not change: I think there was too small of a margin shift to have any meaningful benefit to the change in stock price, as it only went up $0.01 instead of my predicted $1.40
- Evidence path: `docs/locked-changed-input-record.md`, commit hashes above and below.
- Engine note: run on the engine as committed at 7ae0841, before the opioid obligation ($773.8m) was added to the bridge. That item subtracts a constant ≈$1.59/share from both before and after, so the change measured here is unaffected.

Assumption-Challenge Record — Project 1 (ACI)

One row per load-bearing driver: **value · basis · challenge · evidence that would change it.** Base values were proposed by Claude while building the engine and reviewed and accepted by Max in the driver panel on Oct 19, 2026 (see the AI-use register). Partner challenges from the Weeks 5–6 labs are credited to the challenger.

Driver	Value (base)	Basis	Challenge	Evidence that would change it	One-way value swing (band)
FY2030 SG&A / gross profit	94.0% (FY26) → 93.0% (FY30)	FY2025 93.4% ex-opioid charge (10-K); H1 FY2026 adj. EBITDA down 8%; ACI Edge savings unproven	Why assume any improvement when guidance was withdrawn? *(Lab 12 partner: "If the first-quarter decline is your reason for starting low, why do you allow recovery at all?")*	Q3/Q4 FY2026 SG&A rate and adj. EBITDA margin; quantified savings in the Q4 turnaround plan. Flat 94.0% → $12.69	$2.28–$31.41
WACC	7.7% (computed 7.68%)	Rf 5.10% (FRED, Oct 15, 2026); grocery unlevered beta 0.85 (Damodaran, Jan 2026) relevered to 1.78; Kd 5.9% pretax; market weights	Raw beta 0.23 would give a far lower cost of equity; the Lab 06 judgment was 8.0%	A bond-market cost of debt above 5.9%, or a credit-rating change	$8.64–$23.29
Terminal growth	2.0%	Below long-run nominal GDP; mature grocer	Is 2% too high for a chain with negative identical sales?	Identical sales negative through FY2027 → use 1.0–1.5%	$13.11–$20.97
Capex	$2,000m / yr	FY2023–25 $2,156.7m / $1,927.5m / $1,833.6m; FY2026 plan $2.0–2.2bn (10-K)	Turnaround could cut capex, raising FCFF short term but risking store competitiveness	Capex guidance in the Q4 turnaround plan	$14.67–$21.11
FY2026 SG&A / gross profit	94.0%	H1 FY2026 adj. EBITDA $1,797m vs $1,959m prior year; model FY26 EBITDA within 3.4% of the H1 run-rate	Could be worse given Q2 margin of 4.1%	Q3 FY2026 adj. EBITDA margin below 4.0%	$16.72–$19.99
Gross margin	26.8% (FY26) → 27.0% (FY30)	27.8% / 27.7% / 27.2% FY23–25; 26.6% Q1, 26.8% Q2 FY26; pharmacy and digital mix pressure	*(Lab 12 partner: "Trace the 0.5-point gross-margin sensitivity all the way to the value.")*	Gross margin ex-fuel and LIFO trend in Q3/Q4 FY2026	$17.00–$19.22 (FY30)
Revenue growth	−0.5% (FY26), 1.5% (FY28–30)	Identical sales +3.0/+2.0/+2.0% FY23–25, −0.8% Q1 and Q2 FY26; 53rd week removed from base	*(Lab 12 partner: "You started at 0.5% growth even though ACI reported 3.5% FY2025 growth.")* Answer: 3.5% includes the 53rd week; identical sales are the clean measure	Identical sales turning positive for two quarters	FY26 $17.32–$19.32; FY28–30 $16.63–$18.75
Payables	24.2 days of cost of sales	10-K accounts payable $4,021.2m on $60,565.8m cost of sales (AI proposal, verified by Max on the 10-K)	Supplier terms could tighten in a turnaround	Payables days in the next 10-Q/10-K	Not a slider (locked)
Opioid obligation	$773.8m debt-like	10-K Note 12: $655.1m + $21.7m + ≈$97.0m, paid over ≈9 years; undiscounted, pre-tax	Undiscounted and pre-tax overstates the claim slightly	Note 12 updates; payments made; participation levels	−$1.59/share vs ignoring it
Share count	488.2m diluted	Q2 FY2026 diluted weighted-average (8-K, Oct 13, 2026); cross-checks 485.3m cover count, 497.0m adjusted diluted	Buybacks continue; RSUs add 8.8m	Q2 10-Q cover-page count	≈1.8% per-share effect (497.0m)

Causal chains for the main drivers (through the statements)

- **SG&A / gross profit ↑** → SG&A ↑ → EBIT ↓ → NOPAT ↓ → FCFF ↓ every year → terminal FCFF ↓ → EV ↓; the bridge items are fixed, so equity value falls one-for-one with EV. Revenue, inventory and payables do not move.
- **WACC ↑** → every FCFF and the terminal value are discounted harder, and the Gordon spread (WACC − g) widens → EV ↓. No statement line changes.
- **Revenue growth ↑** → revenue ↑ → gross profit and EBIT ↑ at the same ratios; inventory ↑ is now mostly funded by payables ↑ (24.2 vs 31.2 days) → small ΔNWC drag → FCFF ↑ → value ↑ modestly.
- **FY2026 gross margin** only changes FY2026 (FY2030 is set by the long-run margin), so it barely moves value; this is why the Locked Changed-Input Record showed +$0.01.

Does the ranking depend on the ranges?

Yes. On historical bands, FY2030 SG&A/GP leads ($29.13 swing), then WACC ($14.65). With narrow, management-credible bands (SG&A/GP 92.5–93.5%, WACC 7.2–8.2%, g 1.5–2.5%), WACC leads ($6.82) and SG&A/GP falls to fourth ($5.20). The ranking is therefore reported as conditional on the chosen ranges, not as a probability claim (`outputs/report.md`).

Cold-run log — Project 1 (ACI)

Goal: one end-to-end run in a fresh environment, following only the README's "Setup and run" commands, recorded.

Item	Detail
Date / machine	Oct 19, 2026; Windows 11 laptop; PowerShell; Python 3.12.10
Source	Fresh `git clone --branch PROJECT-1` of github.com/mpuntillo33/ticker-research (commit `78a4938`) into a new folder
Environment	New virtual environment (`py -3.12 -m venv .venv`), packages from `requirements.txt` only
Recordings	Attempt 1: 20:28:51, 1:48 long. Attempt 2: 20:34:27, 4:31 long. Kept outside the repo (file size): `Project 1\Submission\cold-run\`

What happened

1. **First failure (attempt 1, 0:00–0:20):** `cd $HOME\Desktop` failed with "*Cannot find path 'C:\Users\maxpu\Desktop' because it does not exist.*" The Desktop on this machine is redirected (OneDrive), so that folder is not at the default path. **Resolution:** ran the README commands from the home folder instead (`C:\Users\maxpu`). This step was a convenience line given in chat, not a README instruction; the README itself does not depend on a Desktop folder.
2. Attempt 1 cloned the repo, created `.venv`, and started `pip install -r requirements.txt`; the recording stopped during the install (no error on screen).
3. **Attempt 2** continued in the same fresh clone and `.venv` (pip reported packages "already satisfied" from attempt 1), then:
   - `tests\test_engine.py`: **10/10 PASS** (known answers $27.4974 and $291.75, articulation, monotonicity, bridge, changed-input, fail-loudly, comps consistency).
   - `analysis.py`: base **$17.89/share**, EV $18,553m, terminal value 79.7%, all checks PASS.
   - `comps.py`: EV/EBITDA ACI 4.31x, KR 6.33x, SFM 6.95x; consistency checks PASS.
   - `tools\make_report.py` and `jupyter lab app.ipynb`: report regenerated; notebook server and kernel started (20:37:25).

Limitations

- The two recordings together cover the run; attempt 2 reused attempt 1's fresh clone and environment rather than re-cloning.
- JupyterLab warned "Notebook app.ipynb is not trusted" (expected for a freshly cloned notebook) and "Could not determine jupyterlab build status without nodejs" (cosmetic).